Juncheng Zhu

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Education

University of Waterloo — Waterloo, ON
Candidate for Honours Computer Science, Co-op — *Sept. 2025 – Present*

- **Cumulative GPA: 90%** (Term 1A)
- **Scholarships: René Descartes National Scholarship ($25k)**, President's Scholarship of Distinction.
- **Relevant Coursework:** Designing Functional Programs (**90%**), Calculus I (**92%**), Algebra (**98%**).

Technical Skills

Languages: Python, C/C++ (STL, Policy-Based DS), TypeScript, JavaScript (ES6+), Java, SQL
Frontend: React, Next.js, HTML5 Canvas API, Tailwind CSS, Chart.js, HTML/CSS
Backend/Data: PyTorch, Node.js, FastAPI, Flask, NumPy, Pandas, PyQt5, SQLite, SQLAlchemy, Streamlit
DevOps/Tools: Git, Docker, DigitalOcean, Vercel, Postman, Linux/Bash, WSL, LaTeX

Projects

Sakura Rain Extension | *JavaScript, HTML5 Canvas, React.js, Chrome Extension API* — Jan. 2026

- Engineered a high-performance Chrome Extension rendering procedural particle physics using **HTML5 Canvas** and **Vanilla JS**.
- Implemented complex geometry rendering using **Cubic Bezier Curves** to simulate organic petal shapes without static assets, ensuring scalability and minimal memory cost.
- Optimized rendering loop with `requestAnimationFrame` and batch processing, maintaining **60 FPS** with 1500+ concurrent particles.
- Built a **React.js** popup UI for real-time adjustments of physics parameters like gravity and wind.

PyFinance Tracker | *Python, Flask, SQLite, Chart.js, Jinja2* — Oct. 2025 – Present

- Developed a secure full-stack financial dashboard using **Flask** and **SQLite**, featuring robust user authentication (login/signup) and session management.
- Integrated **Chart.js** to render interactive data visualizations, enabling users to analyze spending trends via dynamic histograms and pie charts.
- Implemented efficient data filtering and sorting algorithms on the backend, ensuring data integrity through strict input validation and defensive programming.

Deep Q-Learning Snake AI | *Python, PyTorch, Reinforcement Learning, Matplotlib* — Dec. 2025

- Developed an autonomous agent using **Deep Q-Networks (DQN)** to master the classic Snake game.
- Designed a custom **Reward Shaping** mechanism using Manhattan Distance to solve the sparse reward problem and accelerate convergence.
- Analyzed **Model Collapse** phenomena during training and implemented **Epsilon Decay** strategies to balance exploration and exploitation phases.
- Visualized training metrics using Matplotlib to diagnose local optima traps and optimize hyperparameters.

Honours & Awards

USACO Gold Division (Algorithm Design)
Euclid Contest: Honour Roll (95/100, Top 0.1%)
AMC 12: Honor Roll of Distinction (132/150, Top 1%)
Fermat Contest: National Champion (Full Marks)
AP Exams: Perfect Score (5/5) in 10 Subjects
CMO Qualifier (Top 50 in Canada)
IMMC (Modeling): National 2nd Place (2025)
CCC: Honour Roll (66/75, Top 2.5%)
AIME: Distinction (Score: 11/15)
CSMC: Honour Roll (59/60, Top 0.1%)

Experience

Math Club President — Richmond Hill, ON
Bayview Secondary School — *Sept. 2024 – June 2025*

- Led the school's mathematics community, organized weekly training, and mentored members for Euclid/CSMC contests.
- Served as **Team Leader** for HMMT 2023/2024; Achieved **13th Place** in Team Guts Round (2024).